

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 1.2: Ecology — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 1.2: Ecology
Driving Question	DRIVING QUESTION: Why are fish disappearing from a water body that once supported many fish, and what can the community do to restore the ecosystem?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Where Have the Fish Gone? Launching Our Investigation into Ecosystems	Learners viewed photographs or a short clip of a Kenyan river or lake (e.g. Nairobi River, Lake Victoria shoreline near Kisumu) that once had many fish but now has few, green water and rubbish. Teacher note: Before showing the images, ask learners to predict what they will see and why fish may be absent. Allow 30 seconds of WAIT TIME, then cold-call 3 learners. Expect observations such as green scum, litter, a bad smell, dead fish and fishers with empty nets.	An ecosystem is a community of living organisms (biotic) interacting with each other and with non-living (abiotic) factors in an area. Key terms: habitat, population, community, niche, biotic and abiotic factors. Teacher note: Check that learners can name biotic parts (fish, algae, water weeds, bacteria) and abiotic parts (water, temperature, light, oxygen, pH) of the river. A common misconception is that an ecosystem means only the living things. Correct it by having learners sort a list of river components into biotic and abiotic.	Fish disappearance is a symptom that something in the whole ecosystem has changed. Learners can say that fish depend on other organisms and on abiotic conditions, so a change in one part affects the others. Teacher note: Accept initial ideas such as pollution, overfishing and drought as hypotheses, not final answers. Record questions on the Driving Question Board (DQB), for example 'What is killing the fish?' and 'How do we count the fish?'. Learners draw a first draft model of the river ecosystem showing arrows of interaction, to be revised in later lessons.
Lesson 2 Who Lives Where? Habitat Classification and Adaptations in a Kenyan River and Beyond	Learners examined pictures, specimens or a gallery walk of organisms from different Kenyan habitats: aquatic (tilapia, Nile perch, water hyacinth), terrestrial (acacia in the Rift Valley, cactus in Turkana) and aerial or arboreal (birds, bats). Teacher note: Use the Think-Pair-Share routine. Ask 'What features help this organism survive where it lives?' and allow 1 minute of WAIT TIME. Expect learners to notice gills, fins, streamlined bodies, waxy leaves, spines	Organisms are classified as aquatic, terrestrial or aerial according to habitat. Each group has structural and physiological adaptations: for example, fish have gills, fins, scales and a streamlined body; xerophytes have a waxy cuticle, reduced leaves and deep roots; hydrophytes have air spaces and stomata on the upper surface. Teacher note: Stress that adaptations are inherited traits that develop over generations, not changes an individual	Fish are adapted to a narrow range of conditions, such as dissolved oxygen, temperature and clean water. If the habitat changes through pollution, their adaptations no longer help them survive and they die or leave. Teacher note: Link this to the anchoring phenomenon by asking 'Which adaptations of the fish become useless when oxygen drops?' Add questions to the DQB such as 'How many fish live in the river?' Learners add the habitat and

	and wings.	makes on purpose. Use the question 'Would a tilapia survive in a dry habitat? Why not?' to check understanding.	adaptation details to their river model.
Lesson 3 Counting Life: Characteristics of Populations and Direct Counting in the School Compound	Learners went outside to count organisms directly in a marked area of the school compound, such as trees, anthills, goats or flowering plants, and recorded the counts and the area measured. Teacher note: Organise groups of 4 with defined roles (counter, recorder, measurer, checker). Ask each group to predict the count first. Expect differences between groups, which opens a discussion on counting errors and on organisms that move or are hidden.	A population is all the individuals of one species in an area. Population density is the number of individuals per unit area (number / area). Direct counting works for large, visible, slow-moving or stationary organisms in a small area. Teacher note: Model a sample calculation, for example 45 plants in 15 m ² gives a density of 3 plants per m ² . Check units. Limits: direct counting is impractical for small, fast, numerous or hidden organisms such as fish.	Direct counting cannot be used to find the number of fish in a river because they are underwater, move about and are too numerous, so other methods are needed. Learners can say that density lets us compare populations of different areas and times. Teacher note: Ask 'How could the community know if the fish population has dropped?' Record on the DQB: 'How do scientists estimate numbers of organisms that cannot be counted directly?' Learners add a population box with density to their model.
Lesson 4 Quadrat Sampling: Is the Distribution of Grass and Weeds Uniform Across the School Field?	Learners placed quadrats (for example 1 m x 1 m) randomly on the school field, counted the individual plants of selected species (grass, weeds such as Mexican marigold or oxalis) and recorded the numbers in a table. Teacher note: Demonstrate random placement, for example by throwing the quadrat over the shoulder or using random coordinates, to avoid bias. Ask groups to predict whether the numbers will be similar. Expect variation between quadrats, with some densely covered and others bare.	Plants are not spread evenly in a habitat. Quadrat sampling estimates population size: mean number per quadrat divided by quadrat area gives density, and density multiplied by total area gives estimated population size. Teacher note: Work an example, e.g. a mean of 8 plants per 1 m ² quadrat over a field of 400 m ² gives about 3,200 plants. Stress that more quadrats give a more reliable estimate. Common errors are miscounting plants on the quadrat edge and forgetting to multiply by the total area.	Distribution is uneven because of differences in factors like light, water, soil and trampling, so one sample cannot represent the whole habitat. Learners should justify using many random samples. Teacher note: Ask 'Why would a single quadrat mislead the community about the fish population?' Use Claim-Evidence-Reasoning. Add DQB questions such as 'What causes the differences in numbers across the habitat?' Learners add sampling to their model as the way to gather evidence about populations.
Lesson 5 Belt Transect Across a Changing Habitat: How Does Plant Population Change as Conditions Change?	Learners laid a belt transect (a line or strip) across a changing habitat, for example from a wet area near a stream or drain to dry ground, and recorded plant species and counts at set intervals, together with abiotic measurements such as light, soil moisture or soil type. Teacher note: Give a clear roles list and ensure the interval is consistent (e.g. every 2 m). Ask for a prediction before the work starts. Expect to see learners record different species at different points along the line.	Population size changes across a habitat because conditions change. Abiotic factors (light, water, soil nutrients) and biotic factors (competition, grazing) influence which plants grow and how many. Teacher note: Guide learners to plot a graph of numbers against distance and to describe the pattern before explaining it. Remind learners that correlation shows a pattern but does not prove cause, so ask what else could explain the pattern.	Distribution and abundance of organisms follow environmental gradients. In the river, the fish population will likewise change from place to place, or over time, if oxygen, temperature or food change. Teacher note: Ask 'Which part of the river would have the most fish and why?' Record DQB questions like 'Which factors control population size?' Learners update their model with arrows from abiotic and biotic factors to population size.
Lesson 6 How Many Fish Are Left? Capture-Recapture	Learners simulated capture-recapture using a container of dry beans or maize grains to represent fish. They took a first sample and	Population size can be estimated by capture-recapture using: estimated population = (number initially marked x	Capture-recapture is suitable for mobile animals such as fish in Lake Victoria or a river. Repeating the estimate over time

Simulation to Estimate Population Size	marked it (for example with a marker pen), returned it, mixed, took a second sample and counted the marked individuals. Teacher note: Provide a table for recording and ask groups to predict the population size before calculating. Insist on thorough mixing. Learners can then count the real total to check how accurate their estimate was.	number in second sample) / number recaptured. Teacher note: Work an example, e.g. 50 marked x 40 caught second time / 5 recaptured = 400 fish. Assumptions: marks do not harm or wear off, marked individuals mix fully, no births, deaths or migration in between, and capture is random. Learners often forget to divide by the number recaptured, so check this step.	shows whether a population is declining. Teacher note: Ask 'If the estimate falls from 1,200 to 300 over five years, what does it tell the community?' Learners should note the limits of the method and the importance of repeating it. Add DQB questions such as 'What is causing the decline?' and add the method to the model.
Lesson 7 Factors Affecting Population Size: Why Are the Fish Disappearing?	Learners analysed data, graphs or case studies of fish populations, for example fishing records from Lake Victoria, the introduction of Nile perch, and water quality readings (oxygen, temperature, pH). They identified how the factors changed as the number of fish fell. Teacher note: Provide a data card per group and ask them to underline factors that can reduce numbers. Allow 1 minute of WAIT TIME and cold-call 3 groups for their claims.	Population size is controlled by biotic factors (predation, competition, disease, food) and abiotic factors (temperature, oxygen, light, pH), and by human activities such as overfishing and pollution. Natality and immigration increase the population while mortality and emigration reduce it. Teacher note: Check that learners can classify each factor and distinguish density-dependent factors (competition, disease) from density-independent ones (floods, pollution). Misconception to address: a single cause is rarely responsible.	Fish are disappearing because of several interacting factors, such as overfishing, predators, competition, falling oxygen and polluted water, which increase mortality and reduce reproduction. Teacher note: Use Claim-Evidence-Reasoning and require evidence from the data cards. Ask which factor needs further investigation (low oxygen and the green water). Update the DQB with 'Why does the water turn green and why does oxygen drop?' Learners add factor arrows to their model.
Lesson 8 Nitrogen and Carbon Cycles: How Nutrients Move and Why Algae Boom in Our River	Learners studied diagrams and a short case or demonstration of nutrient enrichment, for example of fertilizer runoff from farms near rivers in Kericho or Kisumu and sewage entering the water, leading to green algal growth. Teacher note: Ask learners to predict what happens to a pond when fertilizer is added, then compare with images of an algal bloom. Allow 30 seconds of WAIT TIME before sharing. Expect observations of green scum, then dead plants and a bad smell.	Decomposers and bacteria recycle nutrients: in the nitrogen cycle, nitrogen fixation, nitrification, assimilation, ammonification and denitrification; in the carbon cycle, photosynthesis, respiration, decomposition and combustion. Too many nutrients cause algal bloom (eutrophication). When algae and weeds die, bacteria decompose them and use up dissolved oxygen. Teacher note: Check that learners can trace the path of nitrogen from fertilizer to algae and from dead algae to bacteria. A common misconception is that more nutrients are always good for an ecosystem.	Excess nutrients from farms, sewage and detergents cause algae to boom, then die and decay, so decomposers use up oxygen and fish suffocate. This explains the green water, smell and disappearing fish. Teacher note: Ask learners to build a causal chain with arrows from nutrients to fish death and share it with a partner. Add DQB questions such as 'What else pollutes the river and how can we fix it?' Learners revise their model to include the nutrient cycles and oxygen.
Lesson 9 Pollution, Garbage Rivers and Restoring Our River: The Final Explanation	Learners examined case studies or images of polluted rivers such as the Nairobi River, with garbage, sewage, industrial effluent and plastics, as well as examples of community clean-up and restoration efforts. They identified the sources of pollution and the effects on organisms. Teacher note:	Pollution adds nutrients and harmful substances to rivers, causing algal bloom, low dissolved oxygen, fish deaths and smell. Restoration involves reducing waste at the source, treating sewage and effluent, planting riparian vegetation, regulating fishing, community clean-ups and public	Fish disappeared because pollution and overfishing changed biotic and abiotic conditions in the river, especially through nutrient enrichment, algal bloom and loss of oxygen, so mortality exceeded reproduction. The community can restore the ecosystem by controlling pollution,

	Use a gallery walk with a prompt for each source. Ask learners to predict which sources affect fish most and give 1 minute of WAIT TIME. Expect mentions of plastics, oils, chemicals and untreated waste.	education, with support from laws and bodies such as NEMA. Teacher note: Ensure learners link each solution to the cause it addresses. Value: responsibility and environmental stewardship.	regulating fishing, cleaning and monitoring the river (using capture-recapture and water tests) and educating members. Teacher note: Learners present a final revised model and a community action plan using Claim-Evidence-Reasoning. Return to the DQB and mark which questions are answered, and which need more investigation.
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ATTRIBUTION AND LICENSING

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